9 Partial Derivatives

9.1 Definition and Properties of Partial Derivatives

Definition 9.1: Partial Derivative

In this course, we will focus on functions of two variables, i.e., $f(x, y)$. For a binary function $f(x, y)$, the partial derivative with respect to x is given by:

$$\frac{\partial f}{\partial x} = \lim_{h \to 0} \frac{f(x + h, y) - f(x, y)}{h}$$

and the partial derivative with respect to y is given by:

$$\frac{\partial f}{\partial y} = \lim_{h \to 0} \frac{f(x, y + h) - f(x, y)}{h}$$

provided these limits exist.

In general, let $f(x_1, x_2, \ldots, x_n)$ be a function of n variables. The **partial derivative** of f with respect to the variable x_i is defined as:

$$\frac{\partial f}{\partial x_i} = \lim_{h \to 0} \frac{f(x_1, x_2, \ldots, x_i + h, \ldots, x_n) - f(x_1, x_2, \ldots, x_i, \ldots, x_n)}{h}$$

provided this limit exists.

Some key points to note about partial derivatives:

- $f_x(a, b)$ is the instantaneous rate of change of f at the point (a, b) in the x-direction.
- $f_x(a, b)$ is the slope of the tangent line to $f(a, b)$ at the point $x = a$.

Computing partial derivatives is similar to computing derivatives in single-variable calculus. The only difference is that we treat all other variables as constants when differentiating with respect to one variable.

Example 9.1 (Partial Derivatives of a Binary Function)**.**

Let $f(x, y) = e^{2x} \cos(y)$.

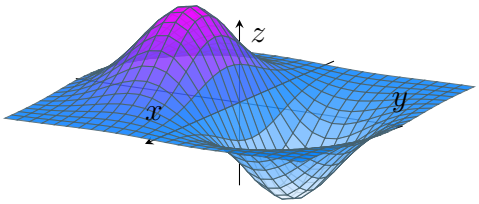
We can now compute the partial derivatives of f with respect to x and y:

$$f_x(x,y) = \frac{\partial}{\partial x}\left(e^{2x}\cos(y)\right) = 2e^{2x}\cos(y)$$
$$f_y(x,y) = \frac{\partial}{\partial y}\left(e^{2x}\cos(y)\right) = -e^{2x}\sin(y)$$

For better comprehension, we can visualize the partial derivatives of a function of two variables as the slopes of tangent lines to the surface defined by the function in the x and y directions.

Example 9.2 (Visualizing Partial Derivatives)**.**

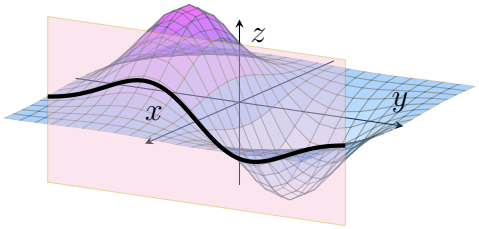
Consider the surface defined by $f(x,y) = -\frac{10y}{e^{x^2+y^2}}$ and its slices along fixed coordinate planes.

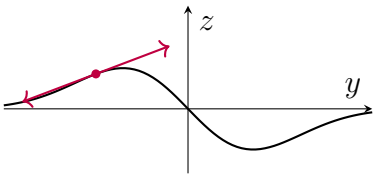


(Original surface $z = f(x,y)$)

$x = 1$ **Trace:**

$$z = f(1,y) = -\frac{10y}{e^{1+y^2}}$$

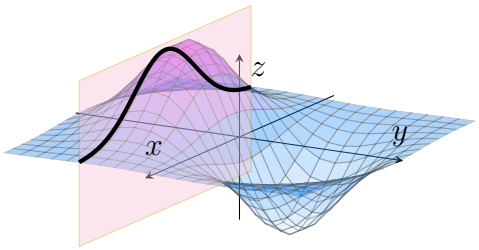




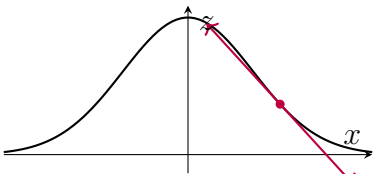
At $(1,-1)$,

$$f_y(1,-1) = \frac{10}{e^2} > 0.$$

$y = -1$ **Trace:**



$$z = f(x, -1) = \frac{10}{e^{x^2+1}}$$

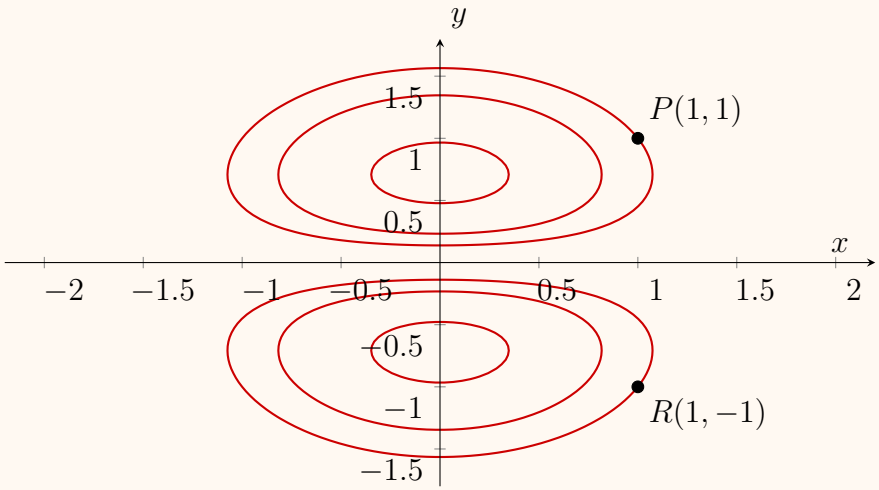


At $(1, -1)$,

$$f_x(1, -1) = -\frac{20}{e^2} < 0.$$

Problem 9.1: Visualizing Partial Derivatives on a Contour Map

Give some reasonable labelings to the contour map based on the image of the graph above.



Determine whether the following values are **positive**, **negative**, or **0**:

1. $f_x(O)$ and $f_y(O)$ at the origin $O(0, 0)$.
2. $f_x(P)$ and $f_y(P)$ at $P(1, 1)$.
3. Compare $|f_x(R)|$ and $|f_y(R)|$ at $R(1, -1)$. Which is greater?

Solution 9.1

1. At $O(0,0)$:
$$f_x(0,0) = 0, f_y(0,0) = -10 < 0$$
2. At $P(1,1)$: As x increases, $f(x,1)$ increases towards 0 from negative values, so $f_x(P) > 0$. At $y = 1$, $f(1,y)$ reaches a local extremum, so $f_y(P) = 0$.
3. At $R(1,-1)$: The contour lines are significantly denser in the y-direction than in the x-direction. Thus:
$$|f_x(R)| > |f_y(R)|$$

9.2 Linear Approximation

Definition 9.2: Linear Function

A **linear function** of two variables is a function of the form

$$f(x,y) = m_x x + m_y y + c$$

The idea of linear approximation is to approximate a function $f(x,y)$ by a linear function $L(x,y)$ that is tangent to the surface defined by f at a given point (a,b).

Given a function $L(x,y) = m_x(x-a) + m_y(y-b) + C$.

Firstly, we want $L(a,b) = f(a,b)$. We then write the linear function as

$$L(x,y) = m_x(x-a) + m_y(y-b) + f(a,b)$$

Secondly, we want the traces of L to be tangent to the traces of f at (a,b). Hence, we have

$$m_x = f_x(a,b), \quad m_y = f_y(a,b)$$

Therefore,

$$L(x,y) = f_x(a,b) \cdot (x-a) + f_y(a,b) \cdot (y-b) + f(a,b)$$

Thus, we now have the following definition of linear approximation:

Definition 9.3: Linear Approximation

The **linear approximation** of a function $f(x, y)$ at the point (a, b) is given by

$$z = f(a, b) + f_x(a, b)(x - a) + f_y(a, b)(y - b),$$

which is the tangent plane.

Problem 9.2: Finding a Linear Approximation

Find the linear approximation of $f(x, y) = \frac{x^2}{y^2+1}$ at $(4, 1)$. Use your answer to approximate $f(4.01, 0.98)$.

Solution 9.2

First, we find the partial derivatives of $f(x, y) = \frac{x^2}{y^2+1}$:

$$f_x(x, y) = \frac{2x}{y^2 + 1}, \quad f_y(x, y) = -\frac{2x^2y}{(y^2 + 1)^2}$$

At the point $(4, 1)$:

$$f(4, 1) = \frac{16}{2} = 8, \quad f_x(4, 1) = \frac{8}{2} = 4, \quad f_y(4, 1) = -\frac{32}{4} = -8$$

The linear approximation is:

$$z = 8 + 4(x - 4) + (-8)(y - 1)$$

Simplifying:

$$z = 8 + 4x - 16 - 8y + 8 = 4x - 8y$$

To approximate $f(4.01, 0.98)$:

$$f(4.01, 0.98) \approx 4(4.01) - 8(0.98) = 16.04 - 7.84 = 8.2$$

9.3 Differentiability

Definition 9.4: Differentiability

A function $f(x, y)$ is said to be **differentiable** at a point (a, b) if the linear approximation of f at (a, b) is a good approximation of f near (a, b).

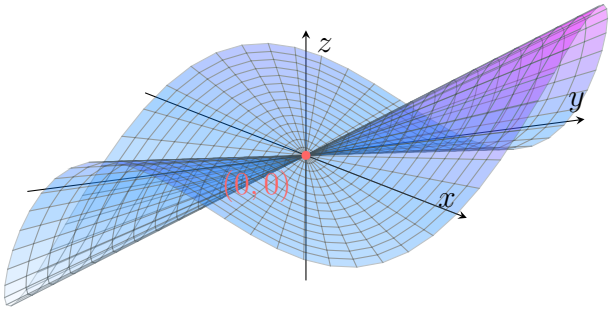
This is equivalent to saying that $f(x, y)$ has a tangent plane at (a, b).

A formal definition of differentiability is as follows:

$$\lim_{(x,y)\to(a,b)} \frac{f(x, y) - L(x, y)}{\sqrt{(x - a)^2 + (y - b)^2}} = 0$$

Example 9.3 (**Non-example:**).

$$g(x, y) = \begin{cases} \dfrac{2xy(x + y)}{x^2 + y^2} & (x, y) \neq (0, 0) \\ \\ 0 & (x, y) = (0, 0) \end{cases}.$$



Note that in single variable calculus, if a function has a derivative at a point, it is also differentiable at that point. However, in multivariable calculus, this is not necessarily the case.

This function $g(x, y)$ is a tricky function. From calculation, we know its partial derivatives,

$\dfrac{\partial g}{\partial x}$ and $\dfrac{\partial g}{\partial y}$, exist at $(0,0)$:

$$\begin{aligned} g_x(0,0) &= \lim_{h\to 0} \frac{g(h,0)-g(0,0)}{h} \\ &= \lim_{h\to 0} \frac{0-0}{h} = 0 \\ g_y(0,0) &= \lim_{h\to 0} \frac{g(0,h)-g(0,0)}{h} \\ &= \lim_{h\to 0} \frac{0-0}{h} = 0 \end{aligned}$$

In fact, from all directions, the derivatives of g exists.

However, g is not **differentiable** at $(0,0)$.

To show this, we can use the definition of differentiability.

$$\begin{aligned} \lim_{(x,y)\to(0,0)} \frac{g(x,y)-L(x,y)}{\sqrt{x^2+y^2}} &= \lim_{(x,y)\to(0,0)} \frac{g(x,y)-g(0,0)+g_x(0,0)(x-0)+g_y(0,0)(y-0)}{\sqrt{x^2+y^2}} \\ &= \lim_{(x,y)\to(0,0)} \frac{g(x,y)}{\sqrt{x^2+y^2}} \\ &= \lim_{(x,y)\to(0,0)} \frac{2xy(x+y)}{(x^2+y^2)\sqrt{x^2+y^2}} \\ &= \lim_{(x,y)\to(0,0)} \frac{2xy(x+y)}{(x^2+y^2)^{3/2}} \end{aligned}$$

We approach $(0,0)$ along the line $y = x$:

$$\begin{aligned}
\lim_{(x,y)\to(0,0)} \frac{2xy(x+y)}{(x^2+y^2)^{3/2}} &= \lim_{x\to 0} \frac{2x \cdot x(x+x)}{(x^2+x^2)^{3/2}} \\
&= \lim_{x\to 0} \frac{2x^2(2x)}{(2x^2)^{3/2}} \\
&= \lim_{x\to 0} \frac{4x^3}{(2x^2)^{3/2}} \\
&= \lim_{x\to 0} \frac{4x^3}{2^{3/2}x^3} \\
&= \lim_{x\to 0} \frac{4}{2^{3/2}} \\
&= \frac{4}{2\sqrt{2}} = \sqrt{2} \neq 0
\end{aligned}$$

This shows that linear approximation is not a good approximation of g near $(0,0)$, and hence g is not differentiable at $(0,0)$.

We now introduce a theorem that provides a sufficient condition for differentiability.

Theorem 9.1: Sufficient Condition for Differentiability

If f_x and f_y are continuous on an open disk D, then $f(x,y)$ is differentiable on D.

9.4 Higher-Order Partial Derivatives

Definition 9.5: Higher-Order Partial Derivatives

We denote by f_{xx} the function

$$f_{xx}(x,y) = \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial x}\right) = \frac{\partial f}{\partial x}$$

and similarly for f_{yy}.

We denote by f_{xy} the function

$$f_{xy}(x,y) = \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial x}\right) = \frac{\partial^2 f}{\partial x \partial y}$$

and similarly for f_{yx}.

Note that the order of differentiation matters for mixed partial derivatives.

The interpretation of higher-order partial derivatives is similar to that of first-order partial derivatives.

- $f_{xx}(a,b)$ is the instantaneous rate of change of f_x at the point (a,b) in the x-direction.
- $f_{xy}(a,b)$ is the instantaneous rate of change of f_x at the point (a,b) in the y-direction.
- $f_{yx}(a,b)$ is the instantaneous rate of change of f_y at the point (a,b) in the x-direction.

Theorem 9.2: Clairaut's Theorem

Suppose that f_{xy} and f_{yx} are continuous on some open disk D. Then

$$f_{xy}(a,b) = f_{yx}(a,b) \quad \text{for all } (a,b) \in D$$

It is similar for functions of more than two variables.

Example 9.4 (Application of Clairaut's Theorem)**.**
Let $g(x,y,z,w) = x^3w^2z^2 + \sin\left(\frac{xy}{z^2}\right)$. Now we want to find g_{zzwx}.

First, we examine the function g. Notice that the first term, $x^3w^2z^2$, is a polynomial function, which is infinitely differentiable over $\mathbb{R}^n$.

The second term, $\sin\left(\frac{xy}{z^2}\right)$, is a composite function. However, sin is infinitely differentiable over $\mathbb{R}$, and $\frac{xy}{z^2}$ also infinitely differentiable over $\mathbb{R}^n, z \neq 0$.

Hence, g is infinitely differentiable over $\mathbb{R}^n, z \neq 0$.

This gives that

$$g_{zzwx} = \frac{\partial g}{\partial z, z, w, x} = g_{wzzx} = g_{xwzz} = \cdots$$

If we calculate this in the original order, we have

$$\begin{aligned} g_{zzwx} &= (((g_z)_z)_w)_x \\ &= (((6x^3w^2z - \frac{2y}{z^3}\cos\left(\frac{xy}{z^2}\right) - \frac{2xy^2}{z^5}\sin\left(\frac{xy}{z^2}\right))_z)_w)_x \\ &= (((6x^3w^2 + \frac{6y}{z^4}\cos\left(\frac{xy}{z^2}\right) + \frac{10xy^2}{z^6}\sin\left(\frac{xy}{z^2}\right) - \frac{4x^2y^3}{z^8}\cos\left(\frac{xy}{z^2}\right))_w)_x \\ &= \cdots \end{aligned}$$

We want to try to avoid this tedious calculation by using Clairaut's Theorem. Rearranging the order of differentiation to g_{xwzz}, we have

$$\begin{aligned} g_{xwzz} &= (((g_x)_w)_z)_z \\ &= (((3x^2w^2z^2 + \frac{y}{z^2}\cos\left(\frac{xy}{z^2}\right))_w)_z)_z \\ &= (((6x^2wz^2 + 0)_z)_z) \\ &= ((12x^2wz + 0)_z) \\ &= 12x^2w \end{aligned}$$

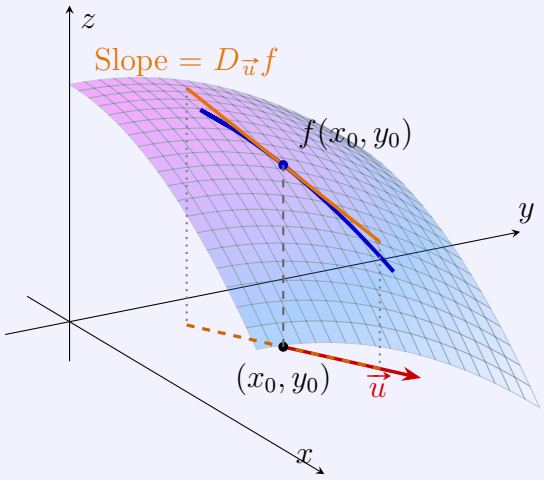
9.5 Directional Derivatives

Definition 9.6: Directional Derivative

Let $\vec{u}$ be a **unit vector**, and let $f(x, y)$ be a function of two variables. The **directional derivative** of f at the point (a, b) in the direction of $\vec{u}$ is defined as

$$D_{\vec{u}}f(\vec{r_0}) = \lim_{t\to 0}\frac{f(\vec{r_0} + t\,\vec{u}) - f(\vec{r_0})}{t}$$

Here, we visualize the directional derivative as the slope of the tangent line to the curve on the surface defined by f in the direction of $\vec{u}$ at the point (a, b).



Example 9.5 (Directional Derivative of a Function of Two Variables)**.**

Let $f(x, y) = ax + by$, $\vec{u} = \langle \cos\theta, \sin\theta \rangle$. We now want to find $D_{\vec{u}} f$.

$$
\begin{aligned}
D_{\vec{u}} f &= \lim_{t \to 0} \frac{f(\vec{r_0} + t\vec{u}) - f(\vec{r_0})}{t} \\
&= \lim_{t \to 0} \frac{f(x_0 + t\cos\theta, y_0 + t\sin\theta) - f(x_0, y_0)}{t} \\
&= \lim_{t \to 0} \frac{a(x_0 + t\cos\theta) + b(y_0 + t\sin\theta) - (ax_0 + by_0)}{t} \\
&= \lim_{t \to 0} \frac{at\cos\theta + bt\sin\theta}{t} \\
&= a\cos\theta + b\sin\theta \\
&= \vec{u} \cdot \langle a, b \rangle
\end{aligned}
$$

9.6 Gradient

Definition 9.7: Gradient

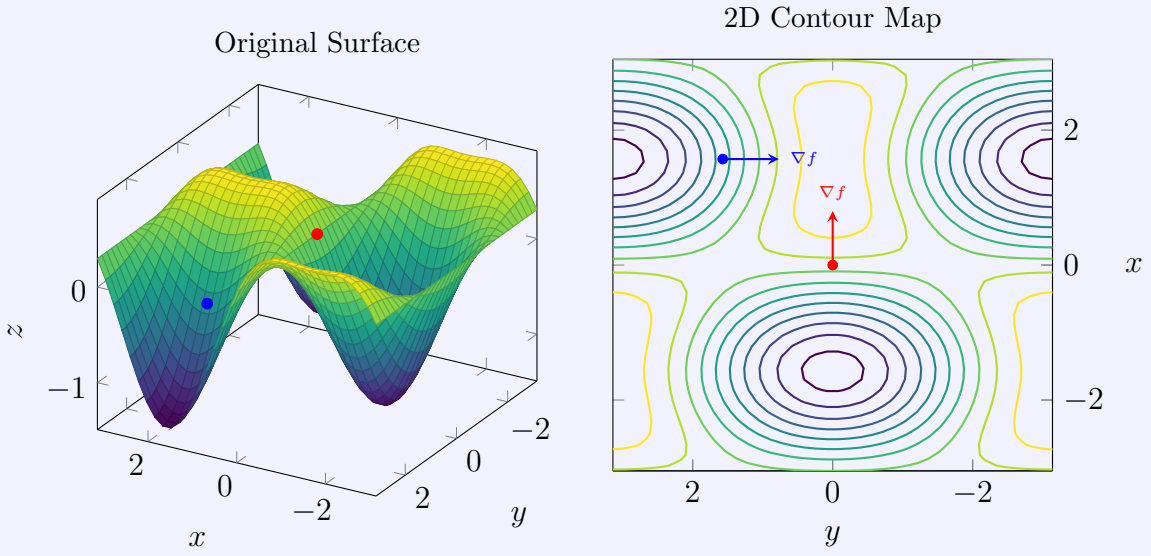
The **gradient** of a function $f(x, y)$ is the vector $\langle a, b \rangle$ such that

$$\langle a, b \rangle = \left\langle \frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right\rangle$$

We call this vector the **gradient vector** of f and denote it by ∇f.

Here, we visualize the gradient vector as a vector that points in the direc-

tion of the steepest ascent of the surface, as shown below in the graphs.



Theorem 9.3: Gradient Theorem

If f is differentiable at (a, b), then

$$\begin{aligned} D_{\vec{u}} f &= \nabla f \cdot \vec{u} \\ &= \|\nabla f\| \cos\theta \end{aligned}$$

Given this, we know that:

- ∇f points in the direction of the **fastest increase** of f.
- $-\nabla f$ points in the direction of the **fastest decrease** of f.
- ∇f is **perpendicular** to the level curves of f.

Example 9.6 (Gradient of a Function of Two Variables)**.**

Let $f(x, y) = \sin(xy^2)$. We have

$$\nabla f = \langle f_x, f_y \rangle = \left\langle y^2 \cos\left(xy^2\right), 2xy \cos\left(xy^2\right) \right\rangle$$

Given this, we can find the gradient of f at $(0, 1)$.

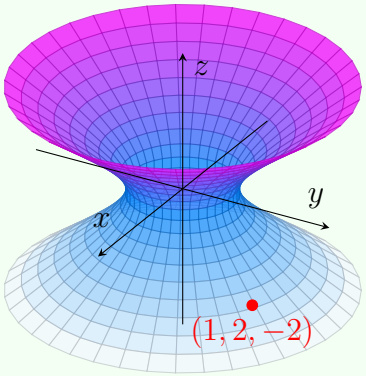
$$\nabla f_{(0,1)} = \langle 1, 0 \rangle$$

Problem 9.3: Finding the Tangent Plane of a Surface

Let $f(x, y, z) = x^2 + y^2 - z^2$, and the surface be defined by $f(x, y, z) = 1$.

Find the tangent plane of the surface at the point $(1, 2, -2)$.

Solution 9.3



First, we compute the gradient of f:

$$\nabla f = \langle f_x, f_y, f_z \rangle = \langle 2x, 2y, -2z \rangle$$

Next, we evaluate the gradient at the point $(1, 2, -2)$:

$$\nabla f(1, 2, -2) = \langle 2, 4, 4 \rangle$$

The equation of the tangent plane at the point (x_0, y_0, z_0) is given by:

$$\nabla f(x_0, y_0, z_0) \cdot \langle x - x_0, y - y_0, z - z_0 \rangle = 0$$

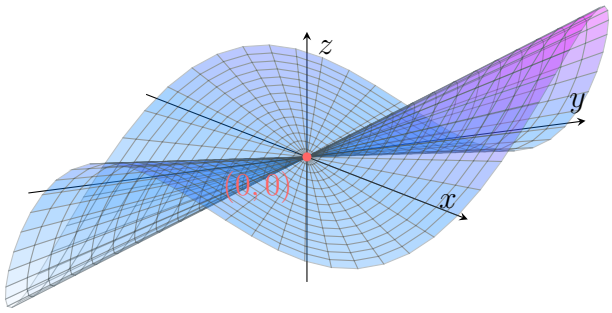
Substituting the values, we have:

$$\begin{aligned}
\langle 2, 4, 4\rangle \cdot \langle x-1, y-2, z+2\rangle &= 0 \\
2(x-1) + 4(y-2) + 4(z+2) &= 0 \\
2x - 2 + 4y - 8 + 4z + 8 &= 0 \\
2x + 4y + 4z - 2 &= 0 \\
x + 2y + 2z - 1 &= 0
\end{aligned}$$

9.7 Differentiability and Continuity

Note that it is dangerous to assume that the existence of partial derivatives implies differentiability.

Recall the example where we proved that the function $g(x, y) = \dfrac{2xy(x+y)}{x^2+y^2}$ is not differentiable at the origin, even though its partial derivatives exist there.



Again, we can compute the gradient of g at $(0,0)$, with $\vec{u} = \langle \cos\theta, \sin\theta \rangle$.

$$\begin{aligned}
D_{\vec{u}} g &= \lim_{t\to 0} \frac{g(\vec{r_0} + t\vec{u}) - g(\vec{r_0})}{t} \\
&= \lim_{t\to 0} \frac{g(t\cos\theta, t\sin\theta) - g(0,0)}{t} \\
&= \lim_{t\to 0} \frac{\dfrac{2(t\cos\theta)(t\sin\theta)(t\cos\theta + t\sin\theta)}{(t\cos\theta)^2 + (t\sin\theta)^2}}{t} \\
&= \lim_{t\to 0} \frac{\dfrac{2t^3(\cos^2\theta\sin\theta + \cos\theta\sin^2\theta)}{t^2(\cos^2\theta + \sin^2\theta)}}{t} \\
&= \lim_{t\to 0} \frac{2t^3(\cos^2\theta\sin\theta + \cos\theta\sin^2\theta)}{t^3} \\
&= \lim_{t\to 0} 2(\cos^2\theta\sin\theta + \cos\theta\sin^2\theta) \\
&= 2(\cos^2\theta\sin\theta + \cos\theta\sin^2\theta) \\
&= 2\cos\theta\sin\theta(\cos\theta + \sin\theta)
\end{aligned}$$

This result is not a constant, and it depends on the direction $\vec{u}$. It yields 0 when $\theta = k\pi/2, k \in \mathbb{Z}$, and ± 1 when $\theta = \pi/4 + k\pi, k \in \mathbb{Z}$.

Note that when

$$D_{\vec{u}} f(x,y) \neq \nabla f \cdot \vec{u}$$

The function is not 'behaving like a plane' at the point (x,y), and hence it is not differentiable at that point.

Theorem 9.4: Differentiability and Continuity

If f_x and f_y are continuous on an open disk D, then f is differentiable on D.

9.8 Chain Rule for Multivariable Functions

Example 9.7 (A Practice Problem)**.**

Given differentiable functions $\vec{r}(t) = \langle x(t), y(t) \rangle$ and $f(x, y)$.

$$\frac{\mathrm{d}}{\mathrm{d}t} f(\vec{r}(t)) = \lim_{h \to 0} \frac{f(\vec{r}(t+h)) - f(\vec{r}(t))}{h}$$

We now compute the linear approximation of f at $\vec{r}(t+h)$:

$$\begin{aligned} f(\vec{r}(t+h)) &\approx f(\vec{r}(t)) + f_x(\vec{r}(t))(x(t+h) - x(t)) + f_y(\vec{r}(t))(y(t+h) - y(t)) \\ &= f(\vec{r}(t)) + f_x(\vec{r}(t))\Delta x + f_y(\vec{r}(t))\Delta y, \\ \text{where } \Delta x &= x(t+h) - x(t), \\ \Delta y &= y(t+h) - y(t) \end{aligned}$$

Hence, the original limit becomes

$$\begin{aligned} \frac{\mathrm{d}}{\mathrm{d}t} f(\vec{r}(t)) &= \lim_{h \to 0} \frac{f(\vec{r}(t)) + f_x(\vec{r}(t))\Delta x + f_y(\vec{r}(t))\Delta y - f(\vec{r}(t))}{h} \\ &= \lim_{h \to 0} \frac{f_x(\vec{r}(t))\Delta x + f_y(\vec{r}(t))\Delta y}{h} \\ &= \lim_{h \to 0} \left(f_x(\vec{r}(t)) \frac{\Delta x}{h} + f_y(\vec{r}(t)) \frac{\Delta y}{h} \right) \\ &= f_x(\vec{r}(t)) \lim_{h \to 0} \frac{\Delta x}{h} + f_y(\vec{r}(t)) \lim_{h \to 0} \frac{\Delta y}{h} \\ &= f_x(\vec{r}(t)) \frac{\mathrm{d}}{\mathrm{d}t} x(t) + f_y(\vec{r}(t)) \frac{\mathrm{d}}{\mathrm{d}t} y(t) \end{aligned}$$

From this, we can derive the Chain Rule for multivariable functions.

Theorem 9.5: Chain Rule for Two-Variable Functions

Let $f(x, y)$ be a differentiable function of two variables, and let $\vec{r}(t) = \langle x(t), y(t) \rangle$ be

a differentiable vector-valued function of one variable. Then

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}t}f(\vec{r}(t)) &= f_x(\vec{r}(t))x'(t) + f_y(\vec{r}(t))y'(t)\\ &= \nabla f(\vec{r}(t)) \cdot \vec{r}'(t)\\ &= \frac{\partial f}{\partial x}\frac{\mathrm{d}}{\mathrm{d}t}x + \frac{\partial f}{\partial y}\frac{\mathrm{d}}{\mathrm{d}t}y\end{aligned}$$

If we examine it from basic concepts, we can see how the Chain Rule naturally holds.

$$\begin{aligned}\frac{\Delta f}{\Delta t} &= \frac{\frac{\Delta f}{\Delta x}\Delta x + \frac{\Delta f}{\Delta y}\Delta y}{\Delta t}\\ &= \frac{\Delta f}{\Delta x}\frac{\Delta x}{\Delta t} + \frac{\Delta f}{\Delta y}\frac{\Delta y}{\Delta t}\end{aligned}$$

From the Chain Rule and the definition of directional derivative, we can derive the following theorem.

Theorem 9.6: Directional Derivative and Gradient

Let $f(x, y)$ be a differentiable function of two variables, and let $\vec{u}$ be a unit vector. Then, $\langle u_1, u_2, D_{\vec{u}}f\rangle$ is tangent to f.

Recall that $D_{\vec{u}}f \perp$ the level curves of f. But why is that? Let's explore this concept further.

If $\vec{r}$ follows a level curve of f, then $f(\vec{r}(t))$ is constant. Hence, $\frac{\mathrm{d}}{\mathrm{d}t}f(\vec{r}(t)) = 0$. Meanwhile, $\frac{\mathrm{d}}{\mathrm{d}t}f(\vec{r}(t)) = \nabla f(\vec{r}(t)) \cdot \vec{r}'(t)$.

Therefore, we have

$$\begin{aligned}&\nabla f(\vec{r}(t)) \cdot \vec{r}'(t) = 0\\ \Longrightarrow\ &\nabla f(\vec{r}(t)) \perp \vec{r}'(t)\end{aligned}$$

Problem 9.4: Partial Derivatives of Composite Functions

Given $x = r\cos\theta, y = r\sin\theta, f(x,y) = x^2 + y^2$, find $\dfrac{\partial f}{\partial r}$ and $\dfrac{\partial f}{\partial \theta}$.

Solution 9.4

We can use the Chain Rule to find the partial derivatives of f with respect to r and θ.

$$\begin{aligned}
\frac{\partial f}{\partial r} &= \frac{\partial f}{\partial x}\frac{\partial x}{\partial r} + \frac{\partial f}{\partial y}\frac{\partial y}{\partial r} \\
&= 2x\cos\theta + 2y\sin\theta \\
&= 2(r\cos\theta)\cos\theta + 2(r\sin\theta)\sin\theta \\
&= 2r(\cos^2\theta + \sin^2\theta) \\
&= 2r
\end{aligned}$$

Similarly, we can find $\frac{\partial f}{\partial \theta}$:

$$\begin{aligned}
\frac{\partial f}{\partial \theta} &= \frac{\partial f}{\partial x}\frac{\partial x}{\partial \theta} + \frac{\partial f}{\partial y}\frac{\partial y}{\partial \theta} \\
&= 2x(-r\sin\theta) - 2y(r\cos\theta) \\
&= -2rx\sin\theta - 2ry\cos\theta \\
&= -2r(r\cos\theta)\sin\theta - 2r(r\sin\theta)\cos\theta \\
&= -2r^2\cos\theta\sin\theta - 2r^2\sin\theta\cos\theta \\
&= -4r^2\cos\theta\sin\theta
\end{aligned}$$

Hereby, we have found the partial derivative for f without directly substituting x and y in terms of r and θ.

Theorem 9.7: Chain Rule for Multivariable Functions

Let $f(x_1, x_2, \ldots, x_n)$ be a differentiable function of n variables, and let $x_i =$

$x_i(t_1, t_2, \ldots, t_m)$ for $i = 1, 2, \ldots, n$. Then,

$$\begin{aligned}\frac{\partial f}{\partial t_j} &= \frac{\partial f}{\partial x_1}\frac{\partial x_1}{\partial t_j} + \frac{\partial f}{\partial x_2}\frac{\partial x_2}{\partial t_j} + \cdots + \frac{\partial f}{\partial x_n}\frac{\partial x_n}{\partial t_j} \\ &= \sum_{i=1}^{n} \frac{\partial f}{\partial x_i}\frac{\partial x_i}{\partial t_j}, \quad j = 1, 2, \ldots, m.\end{aligned}$$